

13 Which suggested mechanism is consistent with the experimentally-obtained rate equation?

	rate equation	suggested mechanism
A	$\text{rate} = k[\text{N}_2\text{O}][\text{H}_2]$	$2\text{NO} + \text{H}_2 \longrightarrow \text{N}_2\text{O} + \text{H}_2\text{O}$ $\text{N}_2\text{O} + \text{H}_2 \xrightarrow{\text{slow}} \text{N}_2 + \text{H}_2\text{O}$
B	$\text{rate} = k[\text{NO}]^2[\text{H}_2]^2$	$2\text{NO} + \text{H}_2 \longrightarrow \text{N}_2\text{O} + \text{H}_2\text{O}$ $\text{N}_2\text{O} + \text{H}_2 \xrightarrow{\text{slow}} \text{N}_2 + \text{H}_2\text{O}$
C	$\text{rate} = k[\text{H}_2\text{O}_2][\text{I}^-]$	$\text{H}_2\text{O}_2 + \text{I}^- \longrightarrow \text{IO}^- + \text{H}_2\text{O}$ $\text{H}_2\text{O}_2 + \text{IO}^- \xrightarrow{\text{slow}} \text{I}^- + \text{H}_2\text{O} + \text{O}_2$
D	$\text{rate} = k[\text{H}_2\text{O}_2][\text{IO}^-]$	$\text{H}_2\text{O}_2 + \text{I}^- \longrightarrow \text{IO}^- + \text{H}_2\text{O}$ $\text{H}_2\text{O}_2 + \text{IO}^- \xrightarrow{\text{slow}} \text{I}^- + \text{H}_2\text{O} + \text{O}_2$

14 D is consistent with the following rate equation: $\text{rate} = k[\text{H}_2\text{O}_2][\text{I}^-]$